Final Design Report

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EK301 Engineering Mechanics I
Section A2
Spring 2026
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Introduction

We wanted to design and construct a truss that maximizes the load-to-cost ratio while satisfying the given geometric and material constraints. The performance of the truss is expected to be limited primarily by the buckling of compression members, so the design approach focused on reducing member lengths while maintaining structural integrity. A custom geometry was developed and evaluated using computational analysis to determine internal member forces and identify critical members under loading. The final design was selected based on its ability to efficiently distribute forces while minimizing total material length and number of joints, both of which directly impact cost. In particular, careful consideration of the truss geometry allowed for shorter compression members and more uniform force distribution, improving overall performance.

Procedure

The overall design procedure used in the final design was similar to that used in the preliminary design report, but modifications were made based on the performance of the initial design. The preliminary truss was found to be insufficient, as certain members experienced forces that exceeded the expected buckling strength based on the class data. In our preliminary designs, we utilized a symmetric design for the load joint that put a single member at the pin under extreme compression. To address this, the geometry of the truss was adjusted to better distribute the applied load and reduce the force carried by any single member. This resulted in a revised custom geometry where member forces were more balanced, ensuring that no individual member exceeded its predicted buckling capacity at our load of 32 ounces. This resulted in a four-member load joint that better distributed the load across the structure. The final design was selected after iterating on these changes and confirming that it satisfied both the structural and cost constraints.

Analysis

The analysis approach remained consistent with the preliminary design, using equilibrium equations to determine internal member forces and identify the critical compression member. However, the results from the preliminary analysis informed key design changes, as it was observed that some members were predicted to fail at loads below the desired range when compared to the class average buckling data. In the final analysis, the buckling strength of each compression member was evaluated using the provided length-dependent relationship, and the design was verified to ensure that all member forces remained below their corresponding buckling limits. Uncertainty in the buckling strength was incorporated by considering the ±Uf it range, allowing for estimation of nominal, strong, and weak failure loads. This ensured that the final design not only met the predicted performance requirements but also accounted for variability in material behavior.

Results

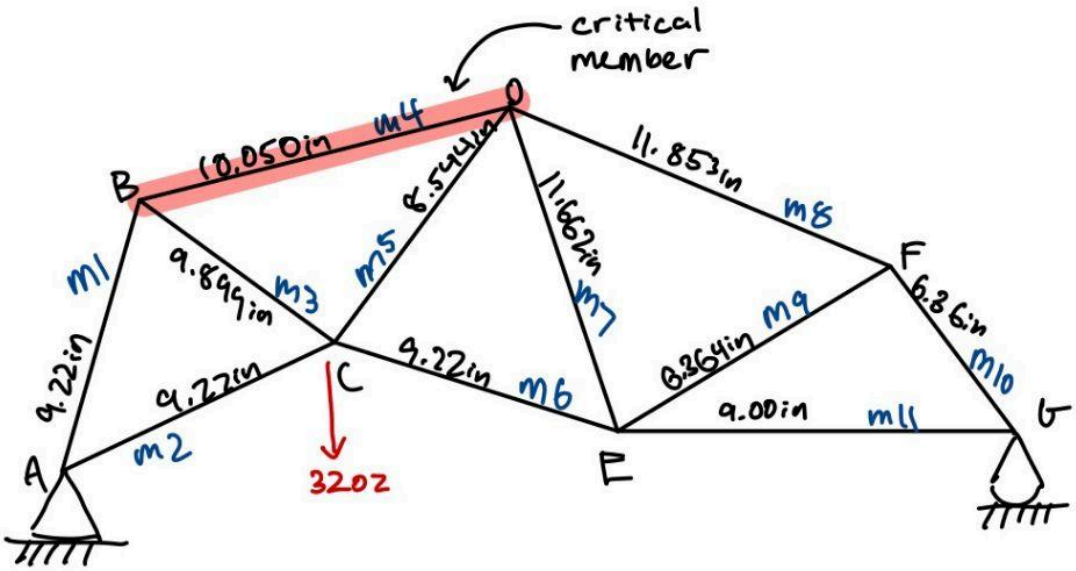


Figure 1. Final Truss Design

Table 1. Final Truss Design Components

Member Number	Member Length (in)	T/C	Buckling Strength (oz)	Uncertainty	Load on Member (oz)
m1	9.220	C	44.113	± 10.00 oz	22.989
m2	9.220	T	--	± 10.00 oz	5.109
m3	9.899	T	--	± 10.00 oz	28.211
m4	10.050	C	37.128	± 10.00 oz	25.059
m5	8.544	T	--	± 10.00 oz	18.437
m6	9.220	T	--	± 10.00 oz	18.912
m7	11.662	C	27.573	± 10.00 oz	8.672
m8	11.853	C	26.692	± 10.00 oz	15.804
m9	6.364	T	--	± 10.00 oz	4.714
m10	6.364	C	92.591	± 10.00 oz	15.085
m11	9.000	T	--	± 10.00 oz	10.667

Code output:

Critical member: m4 (in compression)
Length: 10.050 in
Pcrit Uncertainty: +/- 10.00 oz

Pcrit (nominal): 37.13 oz
Max load (nominal): 47.41 oz

Pcrit (weak): 27.13 oz
Max load (weak): 34.64 oz

Pcrit (strong): 47.13 oz
Max load (strong): 60.18 oz

Cost of truss: $171, Theoretical max load/cost ratio in oz/$: 0.2766

The load is implemented 9 inches away from the pin and two inches above the zero level of the truss, according to project guidelines. The measurements for all of the acrylic was found to be a total of 101.4 inches with seven joints. This leads us to the final cost according to the provided materials of 10$ per joint and 1$ per inch of acrylic material.

Final Cost: $171.40

Dividing our results by our maximum loads (nominal, strong, and weak bound loads), this results in a load to cost ratio for each circumstance. Note that these situations are based on if the acrylic buckling strength was predicted to be weaker or stronger than the collected class average data, which we fit for this analysis. As such, nominal means at the center of uncertainty (+/- 0 oz), weak is the minimum extreme (-10 oz), and strong is the maximum extreme (10 oz.) for the buckling strength of our calculated member. For nominal, we achieved a ratio of 47.41 oz/ $171.40; for strong, 60.18 oz/ $171.40;, and for weak, 34.64 oz./171.40. This results in the following ratio amounts:

Nominal Maximum Load to Cost Ratio: 0.28 oz/ $1 dollar
Strong Maximum Load to Cost Ratio: 0.35 oz/ $1 dollar
Weak Maximum Load to Cost Ratio: 0.20 oz/ $1 dollar

Given that our nominal load should be around our reality, we report our load to cost ratio to be 0.28 oz/ $1 dollar. Our maximum load is nominally 47.41 ounces, and our total truss cost is $171.40 dollars.

Optimization Discussion & Conclusion

The final design that we picked was heavily optimized from our preliminary design. We originally had a symmetrical joint distribution with a lot of symmetry, but it resulted only in a single member under a great amount of compression. As such, we decided to create a joint distribution with four members, leading to the load being more spread out across different members. We utilized the specific equation as below found in the project guidelines:

$$F(L) = C(\frac{Lo}{L})^a \pm Ufit$$

This allowed us to determine our weakest member and understand that we needed to distribute the weight by adding further members to the same joint. We changed the left side of our structure to have a four member joint at the load in order to support the weight better, changing our original plan to have a symmetrical structure towards an asymmetrical, stronger design. We further improved our joint reinforcement strategies. While the tape was able to combine the joints in a strong fashion we quickly realized that the members still needed to be touching in order to avoid erroneous external support forces or wrong interactions.

As we realized that the constraints of this project was about maximizing the load to cost ratio, we used this opportunity to apply this towards our 4-member load joint. Because this was about creating the most efficient truss possible, we reduced our total truss length from our original planned 30 inches down to 27 inches. This cut off a lot of truss that was not supporting as much weight. We further shrunk the maximum height in order to create shorter members, which according to the class collected data, would be much stronger at holding forces. The buckling curve was fit in a way where the longer the member was, the more likely the material was going to buckle at an exponentially smaller force of compression. We thus concentrated our efforts on shortening our members the best we could and ensuring that the load of 32 oz., the required minimum load the truss must hold, is comfortably underneath the minimum buckling load, assuming that all of our acrylic is weak and on the lower bound of the collected buckling data.

Furthermore, we utilized our computer program to assist us in calculating our trusses after testing it vigorously. This allowed us to quickly iterate through multiple designs and understand which ones are viable. This was an incredibly useful tool that allowed us to create a strut to propel our designs forward. With the age of AI, this step in preparation was very insightful and useful, and in future projects, we will utilize this insight to see if we can create, verify, and use our own computerized models to assist us in prototype designs.

There is not much we would change about this overall project. It was a great opportunity to implement our knowledge of truss and force members to create something physical in front of us. On top of that, it gave us a unique opportunity to create an AI tool that can assist us in modeling our designs. This was an important experience in that it allowed us to understand the importance behind modeling and creating these tools for use.

In conclusion, this truss project was an informative experience and an opportunity for us as developing engineers to incorporate AI into a larger-scale project. These types of tasks will only become larger, and the ease of use of our developed AI tool allowed us to tackle this larger project with ease. Keeping in mind the responsibilities of an engineer through the Hartford Roof Collapse case study and the wealth of knowledge and experience we achieved through the design, coding, and building stages of this project, one thing is for certain: we are definitely not the same engineers that started this course in January. We are excited to test our built truss and look forward to implementing this experience further into our careers.

Appendix

Meeting Minutes of Hartford Roof Collapse

Team: 506 Incorporated
Meeting Date: 4/21/2026, 2 hours total Length
Participants: Leihui Tong, Paul Nguyen
Meeting Co-chairs: Paul Nguyen and Leihui Tong
Location: Bay State Road, Private Dormitory for Teamwork
Scribe: Leihui Tong
Prepared Materials: Read and analyzed Hartford Roof Collapse Case Study

Meeting Agenda (Planned):

1. Discuss the Hartford Roof Collapse and understand the failure behind it
2. Discuss how the engineers reacted in the collapse and understand how we need to react as well
3. Explain the current progress on our truss and how to incorporate this study into building it
4. Implement timeline for our project to include error analysis and moderate pre-testing

Important Points Discussed:

Discussed and condensed summary: The Hartford Civic Center arena collapsed in 1978 due to a snowstorm buildup. Originally built to minimize total cost while still building an arena, there were flaws that needed to be addressed and lessons to be learned. We all noted that the collapse of the arena during the night rather than just a couple of hours before, when it was packed with a crowd, was a purely fortunate circumstance. The other outcome was a chilling thought and reinforced the responsibility of our role as engineers to ensure that our products put our clients' safety first, not cost ratios or profits.

We noted primary failure designs. Firstly, the bad structural design where, rather than using a traditional I beam or Tube beam, its + shaped head resulted in a greater susceptibility to buckling. While it was a seemingly newer advancement, this actually contributed to some members being overloaded by 200-800% of its original computer-predicted capacity. Secondly, and most importantly, buckling was not properly accounted for in the computer program that the engineers used. While the analysis gave basic force analysis, the program did not give buckling nodes or buckling force information. As such, when the actual design was implemented, the structure was much weaker in reality compared to the computerized versions. This is a critical step that had us also agree to implement a hand-written analysis of our own project, as well as testing with light loads to simulate the actual load and see if our predictions line up with our computer program's prediction. Lastly, the engineers ignored frequent warning signs, including when multiple large deflections were observed in the beams. Rather than address them, they waived them off as discrepancies that are created when planned models are implemented into reality.

The greatest realization from this discussion was the fact that the engineers on the Hartford arena project trusted their model repeatedly, even when real-world evidence of deviations was presented to them. The constant ignoring of danger in order to implement their cheaper model violated a core principle as an engineer: public safety must come first. The consequences of the collapse, had it been just a couple hours earlier, would have been catastrophic and fatal. We agreed that while our project may not be on the same scale, we would take the same responsibility into our truss to ensure that there would be no catastrophic failures.

We transitioned into talking about our current truss progress. During our preliminary design, we came up with a truss that initially seemed within the boundaries of buckling. While uncertainty could provide the necessary leeway, we quickly scratched our design after learning about this Hartford collapse incident. We would not leave this up to luck and would have our loads be well under buckling to satisfy project requirements. As such, rather than a 3-joint load member, we decided on a 4-member load joint, which allows us to better divide the compressive forces across the members. As we

checked the force distribution across the new design with our computer program, we further decided to utilize hand calculations to double-check the forces. This resulted in equivalent results, heightening our confidence in our design.

We then built the truss according to project standards. We reinforced joints appropriately and taped the supports with generous amounts of material, keeping in mind how the members needed to be joined appropriately in order to function as expected. Rather than do innovative ways that could save on the material, we played it safe according to the Hartford case study, ensuring that our truss would not fail because of reduced effort on the joints.

We wrapped up this discussion by prioritizing completely finishing the truss and to begin testing using ¼ the required weight to see initial buckling behavior. This would confirm our prediction about which member we believe would buckle first, while also allowing us to readjust our computer calculations if this leads to a discovery that proves us wrong. We again promised to not forget the case study incident, and promised to do our best to ensure that our truss not only is predicted to do well, but also does well in reality according to both our calculations and our addressing of any discrepancies that arise.